← Back **Quiz for Week 2**
Graded Assignment • 20 min

Due Aug 29, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. (**Special note:** Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file here. If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)

1 / 1 point

Consider the following linear program

$$\begin{array}{ll} \max & 4x_1 - 2x_2 + x_3 \\ \text{s.t.} & 2x_1 + x_2 \leq 10 \\ & x_2 + x_3 \geq -3 \\ & x_1 + 3x_2 - 3x_3 = 14 \\ & x_1 \geq 0, x_2 \leq 0, x_3 \text{ urs.} \end{array}$$

Let y_1, y_2, and y_3 be the dual variables of the dual linear program, where y_i corresponds to the ith primal constraint. What is the dual constraint corresponding to the primal variable x_2?

- (●) $y_1 + y_2 + 3y_3 \leq -2.$
- ( ) $y_1 + y_2 + 3y_3 \geq 2.$
- ( ) $y_1 + y_2 + 3y_3 \leq 2.$
- ( ) $y_1 + y_2 + 3y_3 \geq -2.$
- ( ) None of the above.

Correct

2. Continue from the previous question. Let $\bar{x}$ and $\bar{y}$ be primal and dual feasible solutions, respectively, where $\bar{y}_i$ corresponds to the ith primal constraint. Which of the following always hold for $\bar{x}$ and $\bar{y}$?

1 / 1 point

- ( ) $4\bar{x}_1 - 2\bar{x}_2 + \bar{x}_3 \geq 10\bar{y}_1 - 3\bar{y}_2 + 14\bar{y}_3.$
- ( ) $4\bar{x}_1 - 2\bar{x}_2 + \bar{x}_3 = 10\bar{y}_1 - 3\bar{y}_2 + 14\bar{y}_3.$
- (●) $4\bar{x}_1 - 2\bar{x}_2 + \bar{x}_3 \leq 10\bar{y}_1 - 3\bar{y}_2 + 14\bar{y}_3.$
- ( ) All of the above.
- ( ) None of the above.

Correct

3. Continue from the previous question. Which of the following always hold for $\bar{x}$ and $\bar{y}$?

1 / 1 point

- ( ) $(10 - 2\bar{x}_1 - \bar{x}_2)\bar{y}_1 = 0.$
- ( ) $(-3 - \bar{x}_2 - \bar{x}_3)\bar{y}_2 = 0.$
- (●) $(14 - \bar{x}_1 - 3\bar{x}_2 + 3\bar{x}_3)\bar{y}_3 = 0.$
- ( ) All of the above.
- ( ) None of the above.

Correct

4. Continue from the previous question. Let x^* and y^* be primal and dual optimal solutions, respectively, where y_i^* corresponds to the ith primal constraint. Which of the following always hold for x^* and y^*?

1 / 1 point

- (●) $y_1^* \geq 0.$
- ( ) $y_2^* \geq 0.$
- ( ) $y_3^* \geq 0.$
- ( ) All of the above.
- ( ) None of the above.

Correct

5. Continue from the previous question. Which of the following statements are correct?

1 / 1 point

- ( ) $4x_1^* - 2x_2^* + x_3^* = 10y_1^* - 3y_2^* + 14y_3^*.$
- ( ) The shadow price of the first primal constraint is y_1^*.
- ( ) $(14 - x_1^* - 3x_2^* + 3x_3^*)y_3^* = 0.$
- (●) All of the above.
- ( ) None of the above.

Correct